

1) Key Measure of Success

To determine if the toll reimbursement successfully encourages driver partners to serve both cities, we need a metric that captures **inter-city mobility**.

A strong candidate is the **Cross-City Trip Ratio** (or the percentage of "Multi-City Drivers"). This measures the proportion of active drivers who completed at least one trip in both Gotham and Metropolis within a single shift or day.

- **Why choose this?** Unlike total earnings or total trips, this metric specifically targets the *behavioral change* the experiment aims to induce. If the toll was the barrier, removing it would see a shift from drivers being "exclusive" to "inclusive" of both territories.

2) Experimental Design

Since we are dealing with a bridge connecting two specific cities, we can't easily split the "geographic" area into a traditional A/B test without risking **network effects** (where the behavior of one group influences the other).

a) Implementation

To implement this, we have a few options for how to divide our subjects:

- **User-Level Randomization:** Randomly select 50% of driver partners to receive the toll reimbursement (Treatment) and 50% who do not (Control).
- **Switchback (Time-Series) Design:** Toggle the reimbursement on and off for *all* drivers at specific time intervals (e.g., every 24 hours).

A **Switchback Design** (also known as a Time-Series experiment) is a great choice here because it helps solve the "network effect" problem. In a marketplace like Gotham and Metropolis, if we only gave the toll reimbursement to *half* the drivers, the other half might feel it's unfair, or the behavior of the "reimbursed" group could change the traffic and demand patterns for everyone else, muddling our data.

Instead of splitting the drivers, we split **time**. We treat the entire geographic area as one unit and toggle the reimbursement on and off across different time windows.

How we could structure this:

- **Unit of Randomization:** Time windows (e.g., 4-hour or 6-hour blocks).
- **The Switch:** During "Treatment" windows, all drivers crossing the bridge get reimbursed. During "Control" windows, no one does.

- **Balancing:** We ensure that both Gotham's busy nights and Metropolis's busy days are represented in both the Treatment and Control groups to account for those circadian rhythms.

Analysis

Once we collect the data from these windows, we need to prove that any change we see isn't just due to random chance

However, in a **Switchback Design**, we often have to deal with "noise." For example, if it rains during a Treatment window, or if there's a huge concert in Metropolis during a Control window, those external factors might skew our results.

To make our conclusion more robust, we might use **Regression Analysis** (specifically a **Linear Regression** or a **Logistic Regression**) as it allows us to control for those extra variables. This ensures that the change we're seeing is actually because of the toll reimbursement and not just because it was a Friday night.

2c) Interpretation and Recommendations

Once we have our p-value from the statistical test (aiming for the standard $p < 0.05$), we need to advise the city operations team.

Possible outcomes:

- **Case A:** The results are statistically significant, and the cross-city ratio increased by **15%**.
- **Case B:** The results are **not** statistically significant—there was almost no change in driver behavior.

Recommendations for Case A (Success)

If we see a significant increase in cross-city activity, the "win" is a more balanced supply of drivers. However, we must address the side effects:

- **Budgeting:** Is the cost of the reimbursed tolls worth the improvement in service levels (e.g., lower wait times for passengers)?
- **Infrastructure:** Increased traffic can lead to faster road wear and higher congestion. We might recommend "Dynamic Reimbursement"—only offering the toll credit during peak hours when the need for cross-city drivers is highest.

Recommendations for Case B (Neutral)

If behavior didn't change, the toll wasn't the primary barrier.

- **Investigate other frictions:** Drivers might stay in one city because they are more familiar with the streets, or perhaps the "Active" hours in Gotham are so lucrative that they have no incentive to leave, even for free.
- **Cost Savings:** The city can save its budget and look for other ways to incentivize movement, such as "Quest" bonuses for completing trips in both cities.

Important Caveats

Even if the data looks good, there are "hidden" factors to consider:

1. **The "Novelty Effect":** Drivers might cross the bridge just because it's a new perk. We need to ensure the behavior stays consistent over several weeks.
2. **Externalities:** Does increased traffic on the bridge create a bottleneck that actually *increases* delivery or travel times, canceling out the benefit of having more drivers?